

Alveo U200 and U250 Accelerator Cards

User Guide

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Introduction



IMPORTANT! Except where noted, this user guide applies to both the U200 and U250 cards.

The AMD Alveo™ U200/U250 data center accelerator cards are peripheral component interconnect express (PCIe®) Gen3 x16 compliant cards featuring the AMD UltraScale+™ technology. These cards accelerate compute-intensive applications such as machine learning, data analytics, video processing, and more. The Alveo U200/U250 data center accelerator cards are available in passive and active cooling configurations. The following figure shows a passively cooled Alveo U200 accelerator card.

Figure 1: Alveo U200 Data Center Accelerator Card (Passive Cooling)



X23434-070820



CAUTION! The Alveo U200/U250 accelerator card with passive cooling is designed to be installed into a data center server, where controlled air flow provides direct cooling. Due to the card enclosure, switches are not accessible and LEDs are not visible (except for the triple-LED module DS3 that protrudes through the left front end PCIe bracket). The card details in this user guide are provided to aid understanding of the card features. If the cooling enclosure is removed from the card and the card is powered-up, external fan cooling airflow **MUST** be applied to prevent over-temperature shut-down and possible damage to the card electronics. Removing the cooling enclosure voids the board warranty.

For additional information about airflow requirements, see the *Alveo U200 and U250 Data Center Accelerator Cards Data Sheet* ([DS962](#)).

See [Appendix A: Additional Resources and Legal Notices](#) for references to documents, files, and resources relevant to the Alveo U200/U250 accelerator cards.

Accelerator Card Overview

Card Features

Features of the AMD Alveo™ U200/U250 data center accelerator cards are listed in the following table.

Table 1: Alveo U200/U250 Features

Card Component	U200	U250
FPGA	UltraScale+ XCU200-2FSGD2104E	UltraScale+ XCU250-2LFIGD2104E
DDR4	64 gigabyte (GB) 4x DDR4 16 GB	
	2400 mega-transfers per second (MT/s), 64 bit with error correcting code (ECC) DIMM	
	x4/x8 unregistered dual inline memory module (UDIMM) support	
Configuration Options	1 gigabit (Gb) Quad Serial Peripheral Interface (SPI) flash memory	
	Micro-AB (USB) JTAG configuration port	
PCIe	16-lane PCI Express	
	PCIe Integrated Endpoint block connectivity	
	Gen1, 2, or 3 up to x16 and Dual Gen4 x8 compatible	
I2C Bus	✓	✓
Status LEDs	✓	✓
Power Management	Power management with system management bus (SMBus) voltage, current, and temperature monitoring	
External Power Sources	65W PCIe slot functional with PCIe slot power only	
	150 W PCIe slot functional with 110 A max V_{CCINT} current PCIe slot power and 6-pin PCIe AUX power cable connected	
	225 W PCIe slot functional with 160 A max V_{CCINT} current PCIe slot power and 8-pin PCIe AUX power cable connected	
Flash Memory	Onboard reprogrammable flash configuration memory	
UART	Universal asynchronous receiver-transmitter (UART) access through the USB port	
Configuration	UltraScale+ device configurable over USB/JTAG and Quad SPI configuration flash memory	

For card specifications, see *Alveo U200 and U250 Data Center Accelerator Cards Data Sheet (DS962)*.

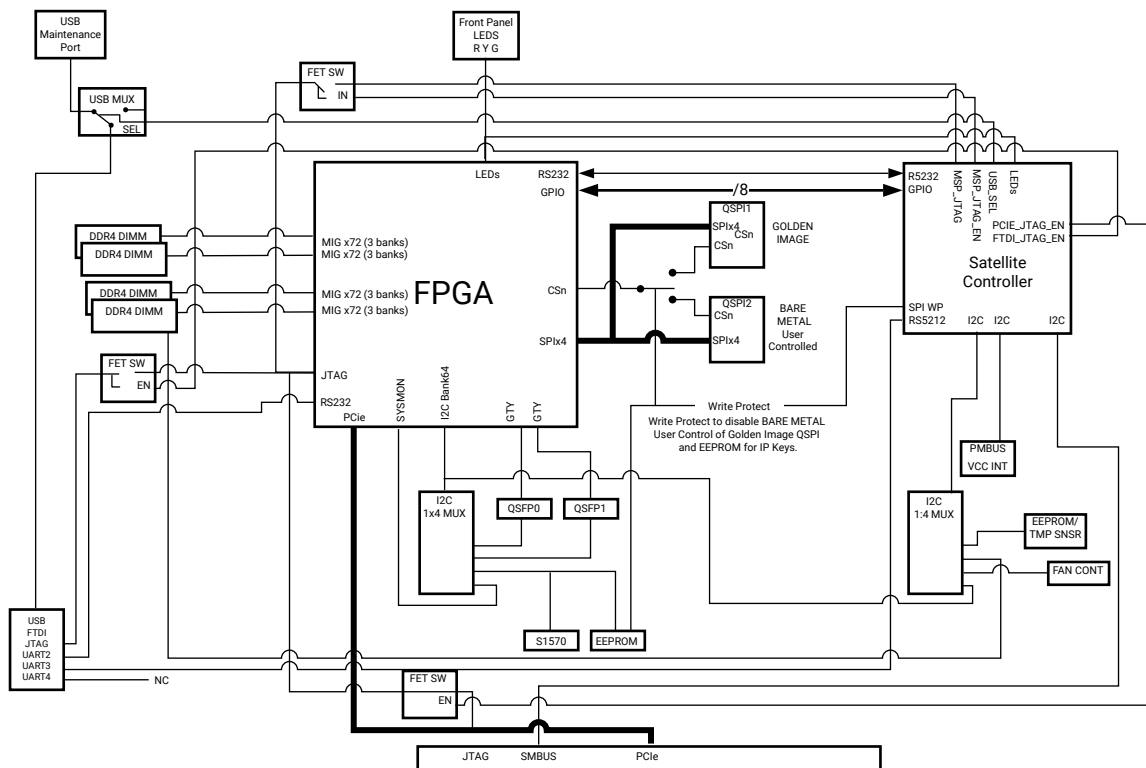
Related Information

Card Component Description

Block Diagram

The block diagram of the Alveo U200/U250 accelerator card is shown in the following figure.

Figure 2: Card Block Diagram



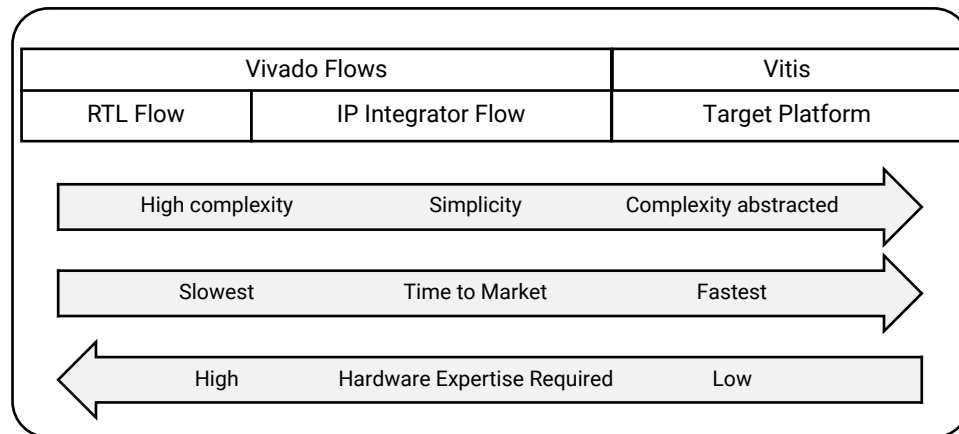
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Note: In the block diagram, QSF0 is port 1, the upper port located on the card PCIe bracket and QSF1 is port 2, the lower port on the PCIe bracket, closest to the PCIe card edge fingers.

Design Flows

The preferred optimal design flow for targeting the Alveo data center accelerator card uses the AMD Vitis™ unified software platform. However, traditional design flows, such as RTL or IP integrator are also supported using the AMD Vivado™ Design Suite. The following figure shows a summary of the design flows.

Figure 3: Alveo Data Center Accelerator Card Design Flows



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Documents related to the different design flows are listed in the following table. Additional details on the Vivado design flow are given in [Chapter 3: Vivado Design Flow](#).

For the Vitis design flow details, see *Vitis Unified Software Platform Documentation* ([UG1416](#)).

Table 2: Documentation to Get Started with Alveo Data Center Accelerator Card Design Flows

	RTL Flow	IP Integrator Flow	Vitis
Flow documentation	UG949 ¹	UG994 ²	UG1416 ³
Vivado tools support	Board support XDC	Vivado Board File	N/A
Programming the FPGA	Vivado Hardware Manager	Vivado Hardware Manager	UG1377 ⁴

Notes:

1. *UltraFast Design Methodology Guide for FPGAs and SoCs* ([UG949](#)).
2. *Vivado Design Suite User Guide: Designing IP Subsystems using IP Integrator* ([UG994](#)).
3. [Vitis Accelerated Software Development Flow Documentation](#) in the Application Acceleration Development flow of the *Vitis Unified Software Platform Documentation* ([UG1416](#)).
4. *Alveo Programming Cable User Guide* ([UG1377](#)).

Vivado Design Flow

This section provides a starting point for expert HDL developers using the RTL and IP integrator flows using the Vivado™ tools.

Vivado Board Support and XDC Files Installation

Prior to creating a Vivado RTL project based on an Alveo™ card, download the Alveo Vivado board support and XDC files associated with your Alveo card from the **Getting Started** tab of the respective Alveo card landing page:

- [Alveo U200 Product Page](#)
- [Alveo U250 Product Page](#)

1. Close all instances of Vivado tools.
2. Navigate to the desired Alveo product page, and download the Vivado board zip file.
3. Update the Vivado board support repository by extracting the Vivado Board ZIP file to the boards directory. For Vivado 2021.1 and greater, the boards directory is located within the AMD Vivado Design Suite installation path at:

```
\data\xhub\boards\XilinxBoardStore\boards\Xilinx\
```

For example, if your Vivado tools installation is located at:

```
C:\Xilinx\Vivado\<version>
```

where <version> is the Vivado tools version you are using, then for Vivado 2021.1 and greater extract the contents to:

```
C:\Xilinx\Vivado\<version>\data\xhub\boards\XilinxBoardStore\boards\Xilinx\
```

For Vivado releases prior to 2021.1, the boards directory is located within the AMD Vivado Design Suite installation path at:

```
\data\boards\board_files
```

4. In addition to the Vivado board file, also download the Xilinx Design Constraints (XDC) file from the Alveo product page. The location of the XDC file on your system is not important.

After you install the Vivado board file and download the XDC file, you can create a Vivado RTL project.

Creating a Vivado RTL Project

There are two flows available for the RTL developer: board aware and XDC based flow. In either flow, you can add RTL files or block designs to your project. The steps necessary for creating a project for each flow are described in the following sections.

Board Aware Flow

Use the following steps to create a board aware Vivado RTL project using the Vivado board files downloaded from the Alveo product page. The Vivado board files associated with the card is used to configure IP and generate constraints for IP used in the block design.

1. Launch Vivado tools.
2. Create a new project by clicking on **File → Project → New**. Click **Next**.
3. Add a project name and click **Next**.
4. Select **RTL Project** as the Project Type and click **Next**.
5. Within the Default Part window, select **Boards**, in the search tab enter the card name you are designing for and click **Next → Finish**. This will create a new RTL project based on the selected accelerator card.

XDC Based Flow

Use the following steps to create an XDC based Vivado RTL project using the XDC downloaded from the Alveo product page. The XDC is a reference for the static pinout of the card.

Within the XDC based Vivado flow, the AMD Alveo™ parts are not visible in the Vivado part selection window. A project using a Vivado part can only be created using a Tcl command. Use the following steps:

1. Launch Vivado tools.
2. In the Tcl console, run the following command:

```
create_project <project> <path> -part <part number>
```

where,

- <project> is the name of the project you want to create

- `<path>` is the path where you want to create the project
- `<part number>` is the Vivado part number as defined in the following table.

Table 3: Vivado Part Number

Alveo Card	Alveo Part	Vivado Part Number
A-U200	XCU200-2FSGD2104E	XCU200-FSGD2104-2-E
A-U250	XCU250-L2FIGD2104E	XCU250-FIGD2104-2L-E

3. Add the card XDC file to the project by clicking on **File → Add Sources → Add or Create Constraints**. Click **Next**. Select **Add Files**. Navigate to the location of the XDC file and click **OK**.

UltraScale+ Device Configuration

The Alveo U200/U250 accelerator card supports two AMD UltraScale+™ FPGA configuration modes:

- Quad SPI flash memory
- JTAG using USB JTAG configuration port

The FPGA bank 0 mode pins are hardwired to master SPI mode $M[2:0] = 001$ with pull-up/down resistors.

At power up, the FPGA is configured by the Quad SPI NOR flash device using the primary serial configuration mode. Refer to the XDC for recommended configuration parameters specified via the various `BITSTREAM.CONFIG` constraints.

If the JTAG cable is plugged in, QSPI configuration might not occur. JTAG mode is always available independent of the mode pin settings.

For complete details on configuring the FPGA, see the *UltraScale Architecture Configuration User Guide* ([UG570](#)).

Table 4: Configuration Modes

Configuration Mode	M[2:0]	Bus Width	CCLK Direction
Master SPI	001	x1, x2, x4	FPGA output
JTAG	Not applicable – JTAG overrides	x1	Not applicable

MCS File Generation and Alveo Card Programming

This section outlines the steps to generate and program the MCS file.

MCS File Generation

The MCS file represents the PROM image which is loaded onto the Alveo accelerator card at power on. It is generated using the `write_cfgmem` tool. This section outlines the steps to generate and program the MCS file.

Prior to generating the MCS file, ensure your project XDC file sets the following properties.

- `CONFIG_VOLTAGE`
- `BITSTREAM.CONFIG.CONFIGFALLBACK`
- `BITSTREAM.GENERAL.COMPRESS`
- `CONFIG_MODE`
- `BITSTREAM.CONFIG.SPI_BUSWIDTH`
- `BITSTREAM.CONFIG.CONFIGRATE`
- `BITSTREAM.CONFIG.EXTMASTERCLK_EN`
- `BITSTREAM.CONFIG.SPI_FALL_EDGE`
- `BITSTREAM.CONFIG.UNUSEDPIN`
- `BITSTREAM.CONFIG.SPI_32BIT_ADDR`

Use the following command with the parameters outlined in [Table 5](#) to generate the MCS file.

```
write_cfgmem -force -format mcs -interface <interface_type> -size <size> -loadbit "up <user_config_region_offset> <input_file.bit>" -file "<output_file.mcs>"
```

Table 5: write_cfgmem Parameter Settings

write_cfgmem Parameter	Setting
interface_type	spix4
size	128
user_config_region_offset ¹	0x01002000
input_file.bit	Filename of the input .bit file

Table 5: write_cfgmem Parameter Settings (cont'd)

write_cfgmem Parameter	Setting
output_file.mcs	MCS output filename

Notes:

1. Address 0x00000000 through 0x01001FFF is a write protected region which holds the card's golden recovery image and cannot be written to. The user_config_region_offset setting cannot be within this range.

For additional details on write_cfgmem, see the *UltraScale Architecture Configuration User Guide* (UG570).

Program the Alveo Data Center accelerator card

After the MCS file has been generated, use the following steps to flash the FPGA on the Alveo Data Center accelerator card using the Vivado hardware manager. Detailed steps for programming the FPGA are outlined in the chapter Programming the FPGA Device in the *Vivado Design Suite User Guide: Programming and Debugging* (UG908).



RECOMMENDED: Programming through JTAG maintenance port must be from a separate machine to avoid PCIe downlink causing the server to reboot during programming. Alternatively, the PCIe link can be manually disabled through software and rescanned after programming is complete.

1. Connect to the Alveo U200/U250 data center accelerator card using the Vivado hardware manager via the Micro-USB port.
2. Select **Add Configuration Device** and select the mt25qu01g-spi-x1_x2_x4 part.
3. Select OK when prompted "Do you want to program the configuration memory device now?" or right-click the target to select **Program the Configuration Memory Device**.
 - a. Select the MCS file target.
 - b. Select **Configuration File Only**.
 - c. Click **OK**.
4. After programming has completed, disconnect the card in the hardware manager, and disconnect the JTAG programming cable from the Alveo accelerator card.
5. Perform a cold reboot on the host machine to complete the card update.



IMPORTANT! When switching between an Alveo data center accelerator card target platform and a custom design, revert the card to the factory image before loading an alternate image into the PROM. See [Answer Record 71757](#) for more information.

Card Installation

Standard ESD Measures



CAUTION! ESD can damage electronic components when they are improperly handled, and can result in total or intermittent failures. Always follow ESD-prevention procedures when removing and replacing components.

To prevent ESD damage:

- Attach a wrist strap to an unpainted metal surface of your hardware to prevent electrostatic discharge from damaging your hardware.
- When you are using a wrist strap, follow all electrical safety procedures. A wrist strap is for static control. It does not increase or decrease your risk of receiving electric shock when you are using or working on electrical equipment.
- If you do not have a wrist strap, before you remove the product from ESD packaging and installing or replacing hardware, touch an unpainted metal surface of the system for a minimum of five seconds.
- Do not remove the device from the antistatic bag until you are ready to install the device in the system.
- With the device still in its antistatic bag, touch it to the metal frame of the system.
- Grasp cards and boards by the edges. Avoid touching the components and gold connectors on the adapter.
- If you need to lay the device down while it is out of the antistatic bag, lay it on the antistatic bag. Before you pick it up again, touch the antistatic bag and the metal frame of the system at the same time.
- Handle the devices carefully to prevent permanent damage.

Installing Alveo Data Center Accelerator Cards in Server Chassis

For hardware installation procedures, see *Getting Started with Alveo Data Center Accelerator Cards* ([UG1301](#)).

Because each server or PC vendor's hardware is different, for physical board installation guidance, see the manufacturer's PCI Express® board installation instructions.

Card Component Description

This chapter provides a functional description of the components of the Alveo™ U200/U250 data center accelerator card.

UltraScale+ Device

The Alveo U200 accelerator card is populated with the UltraScale+ XCU200-L2FSGD2104E device Vivado part number: XCU200-FSGD2104-2-E.

The Alveo U250 accelerator card is populated with the UltraScale+ XCU250-L2FIGD2104E device Vivado part number: XCU250-FIGD2104-2L-E.

DDR4 DIMM Memory

Four 288-pin DDR4 DIMM sockets are populated with single rank DIMMs capable of operating at data rates up to 2400 MegaTransfers per second (MT/s).

Table 6: DDR4 Interfaces

Alveo Card	Parameter	Description
A-U200-A64G-PQ-G A-U200-P64G-PQ-G A-U250-A64G-PQ-G A-U250-P64G-PQ-G	Manufacturer	Micron
	Part Number	MTA18ASF2G72PZ-2G3B1
	Description	16 GB 288-pin DDR4 RDIMM
		Configuration: 2 Gb x 72
		Single rank
		Supports ECC error detection and correction
		Supports 2400 MT/s

The detailed FPGA and DIMM pin connections are documented in the Alveo U200/U250 accelerator card XDC file found in the **Getting Started** tab of the respective Alveo card landing page:

- [Alveo U200 Product Page](#)

- [Alveo U250 Product Page](#)

For more details about the Micron DDR4 DIMM, see the Micron MTA18ASF2G72PZ-2G3B1IG data sheet at the Micron website: <http://www.micron.com>.

Quad SPI Flash Memory

The Quad SPI device provides 1 Gb of nonvolatile storage.

- Supply voltage: 1.8V
- Datapath width: 4 bits

For flash memory details, see the Micron data sheet at the Micron website.

For configuration details, see the *UltraScale Architecture Configuration User Guide* ([UG570](#)). The detailed FPGA and Flash pin connections for the feature described in this section are documented in the Alveo U200/U250 accelerator card XDC file found in the **Getting Started** tab of the respective Alveo card landing page:

- [Alveo U200 Product Page](#)
- [Alveo U250 Product Page](#)

FT4232HQ USB-JTAG/UART Interface

The FT4232HQ Quad USB-UART provides a UART connection through the micro-AB USB connector. The FPGA UART TX/RX (two-wire) connection is made through the FT4232HQ BD port. Channel BD implements a 2-wire level-shifted TX/RX UART connection to the FPGA. The FTDI FT4232HQ data sheet is available on the FTDI website: <https://www.ftdichip.com/>.

PCI Express Endpoint

The Alveo U200/U250 accelerator card implements a 16-lane PCI Express® edge connector that performs data transfers at the rate of 2.5 giga-transfers per second (GT/s) for Gen1, 5.0 GT/s for Gen2, and 8.0 GT/s for Gen3 applications. The -2 speed grade FPGA included with the cards supports up to Gen3 x16.

The detailed UltraScale+ device and PCIe pin connections for the feature described in this section are documented in the accelerator card design constraints (XDC) file. The endpoint is compliant to the v3.0 specification.

QSFP28 Module Connector

The Alveo accelerator cards host two 4-lane small form-factor pluggable (QSFP) connectors that accept an array of optical modules. Each connector is housed within a single QSFP cage assembly. Note QSFP0 is port 1, the upper port located on the PCIe bracket and QSFP1 is port 2, the lower port located on the PCIe bracket, closest to the PCIe card edge fingers.

Refer to the section for the FPGA clock sources. The detailed device and QSFP pin connections for the feature described in this section are documented in the XDC file found in the **Getting Started** tab of the respective product page:

- [Alveo U200 Product Page](#)
- [Alveo U250 Product Page](#)

The QSFP+ connectors are accessible via the I2C interface on the Alveo U200/U250 accelerator cards. The QSFP connector's sideband signals are accessible directly from the FPGA. The MODSELL, RESETL, MODPRSL, INTL, and LPMODE sideband signals are defined in the following small form factor (SFF) specifications.

For additional information about the quad SFF pluggable (28 Gb/s QSFP+) module, see the SFF-8663 and SFF-8679 specifications for the 28 Gb/s QSFP+ at the SNIA Technology Affiliates website: <https://www.snia.org/sff/specifications2>.

QSFP0

The QSFP0 interface has the high-speed connections to the UltraScale+ device listed in the following table.

Table 7: QSFP0 Interface High-Speed Connections

Name	Direction	Bank	I/O Reference	Pin	Description
MGT_SI570_CLOCK0_C_N	Input	231	MGTREFCLK0N_231	M10	REFCLK0 Negative LVDS Signal
MGT_SI570_CLOCK0_C_P	Input	231	MGTREFCLK0P_231	M11	REFCLK0 Positive LVDS Signal
QSFP0_CLOCK_N	Input	231	MGTREFCLK1N_231	K10	REFCLK1 Negative LVDS Signal
QSFP0_CLOCK_P	Input	231	MGTREFCLK1P_231	K11	REFCLK1 Positive LVDS Signal
QSFP0_RX1_N	Input	231	MGTYRXN0_231	N3	Negative Receive Lane 1 Input

Table 7: QSFP0 Interface High-Speed Connections (cont'd)

Name	Direction	Bank	I/O Reference	Pin	Description
QSFP0_RX2_N	Input	231	MGTYRXN1_231	M1	Negative Receive Lane 2 Input
QSFP0_RX3_N	Input	231	MGTYRXN2_231	L3	Negative Receive Lane 3 Input
QSFP0_RX4_N	Input	231	MGTYRXN3_231	K1	Negative Receive Lane 4 Input
QSFP0_RX1_P	Input	231	MGTYRXP0_231	N4	Positive Receive Lane 1 Input
QSFP0_RX2_P	Input	231	MGTYRXP1_231	M2	Positive Receive Lane 2 Input
QSFP0_RX3_P	Input	232	MGTYRXP2_231	L4	Positive Receive Lane 3 Input
QSFP0_RX4_P	Input	231	MGTYRXP3_231	K2	Positive Receive Lane 4 Input
QSFP0_TX1_N	Output	231	MGTYTXN0_231	N8	Negative Transmit Lane 1 Output
QSFP0_TX2_N	Output	231	MGTYTXN1_231	M6	Negative Transmit Lane 2 Output
QSFP0_TX3_N	Output	231	MGTYTXN2_231	L8	Negative Transmit Lane 3 Output
QSFP0_TX4_N	Output	231	MGTYTXN3_231	K6	Negative Transmit Lane 4 Output
QSFP0_TX1_P	Output	231	MGTYTXP0_231	N9	Positive Transmit Lane 1 Output
QSFP0_TX2_P	Output	231	MGTYTXP1_231	M7	Positive Transmit Lane 2 Output
QSFP0_TX3_P	Output	231	MGTYTXP2_231	L9	Positive Transmit Lane 3 Output
QSFP0_TX4_P	Output	231	MGTYTXP3_231	K7	Positive Transmit Lane 4 Output

QSFP1

The QSFP1 interface has the high-speed connections to the UltraScale+ device listed in the following table.

Table 8: QSFP1 Interface High-Speed Connections

Signal Name	Direction	Bank	I/O Reference	Pin	Description
MGT_SI570_CLOCK1_C_N	Input	230	MGTREFCLK0N_230	T10	REFCLK0 Negative LVDS Signal
MGT_SI570_CLOCK1_C_P	Input	230	MGTREFCLK0P_230	T11	REFCLK0 Positive LVDS Signal
QSFP1_CLOCK_N	Input	230	MGTREFCLK1N_230	P10	REFCLK1 Negative LVDS Signal
QSFP1_CLOCK_P	Input	230	MGTREFCLK1P_230	P11	REFCLK1 Positive LVDS Signal

Table 8: QSFP1 Interface High-Speed Connections (cont'd)

Signal Name	Direction	Bank	I/O Reference	Pin	Description
QSFP1_RX1_N	Input	230	MGTYRXN0_230	U3	Negative Receive Lane 1 Input
QSFP1_RX2_N	Input	230	MGTYRXN1_230	T1	Negative Receive Lane 2 Input
QSFP1_RX3_N	Input	230	MGTYRXN2_230	R3	Negative Receive Lane 3 Input
QSFP1_RX4_N	Input	230	MGTYRXN3_230	P1	Negative Receive Lane 4 Input
QSFP1_RX1_P	Input	230	MGTYRXP0_230	U4	Positive Receive Lane 1 Input
QSFP1_RX2_P	Input	230	MGTYRXP1_230	T2	Positive Receive Lane 2 Input
QSFP1_RX3_P	Input	230	MGTYRXP2_230	R4	Positive Receive Lane 3 Input
QSFP1_RX4_P	Input	230	MGTYRXP3_230	P2	Positive Receive Lane 4 Input
QSFP1_TX1_N	Output	230	MGTYTXN0_230	U8	Negative Transmit Lane 1 Output
QSFP1_TX2_N	Output	230	MGTYTXN1_230	T6	Negative Transmit Lane 2 Output
QSFP1_TX3_N	Output	230	MGTYTXN2_230	R8	Negative Transmit Lane 3 Output
QSFP1_TX4_N	Output	230	MGTYTXN3_230	P6	Negative Transmit Lane 4 Output
QSFP1_TX1_P	Output	230	MGTYTXP0_230	U9	Positive Transmit Lane 1 Output
QSFP1_TX2_P	Output	230	MGTYTXP1_230	T7	Positive Transmit Lane 2 Output
QSFP1_TX3_P	Output	230	MGTYTXP2_230	R9	Positive Transmit Lane 3 Output
QSFP1_TX4_P	Output	230	MGTYTXP3_230	P7	Positive Transmit Lane 4 Output

Related Information

[Clocks](#)

I2C Bus

The Alveo U200/U250 accelerator cards support the SMBus/I2C interface for out-of-band card management. The server Board Management Controller will communicate using SMBus to the satellite controller over the PCIe edge connector. For more information on out-of-band-card management, see [Alveo Card Out-of-Band Management Specification for Server BMC](#).

Status LEDs

The Alveo card operates with the passive heat sink enclosure cover installed, which makes it difficult to see the DS1 and DS2 LEDs. The DS3, D4, and DS5 LEDs are visible through a cutout in the PCIe end bracket. The following table defines the card status LEDs.

Table 9: Card Status LEDs

Reference Designator	Description
DS1	RED: POWER_GOOD When illuminated solid red, the power to the card is not within the specification. During power on, the LED might blink, which is expected behavior.
DS2	BLUE: DONE_0 When illuminated the FPGA is successfully programmed.
DS3	ORANGE: STATUS_LED0 Not defined.
DS4	YELLOW: STATUS_LED1 Not defined.
DS5	GREEN: STATUS_LED2 Not defined.

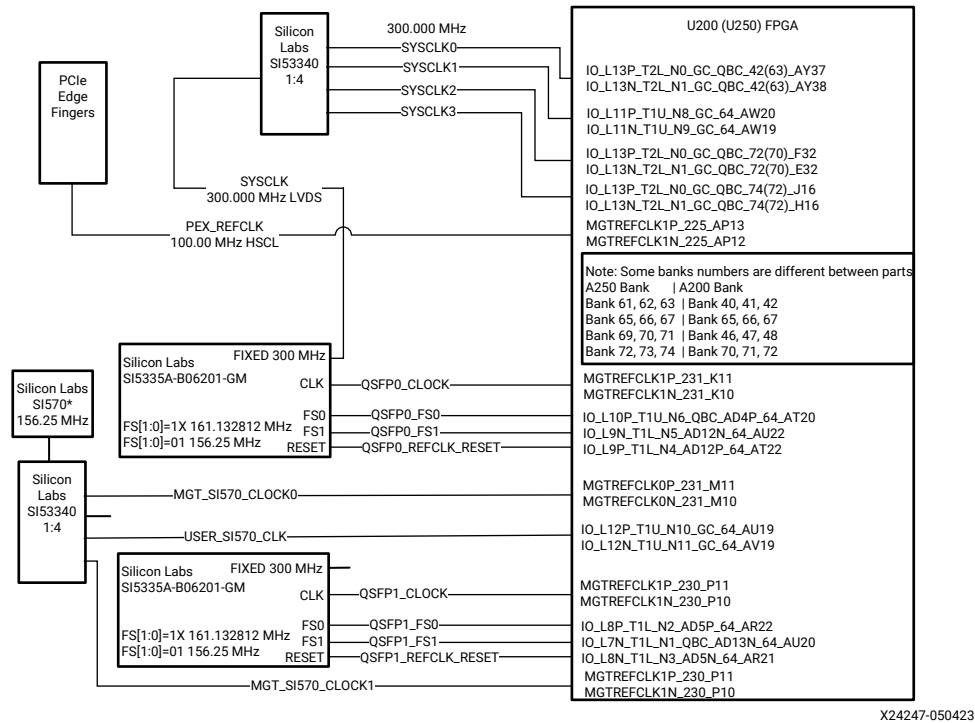
Card Power System

Limited power system telemetry is available through the card management system (CMS) I2C IP. See *Card Management Solution Subsystem Product Guide* ([PG348](#)) for information on how the I2C IP is instantiated in the UltraScale+ IP integrator flow. See [Design Flows](#) for more information.

Clocks

The following figure shows the clocks generated and connected to the FPGA. The detailed FPGA, clock pin connections, and frequencies for the feature described in this section are further documented in the U200/U250 accelerator card XDC file.

Figure 4: Alveo U200/U250 Clocking



Miscellaneous I/O

On the U200 and U250 cards, the UltraScale+ device has a single bank with various inputs and outputs to communicate with the satellite controller and other devices on the card. See the XDC file for all of the signal connections. This section only references the signals that are usable in bank 64. Bank 64 is powered from a 1.2V source. Care needs to be taken when using the LVDS Clocks on a 1.2V bank. A DQS_BIAS needs to be used or another IO standard which re-biases the signals internal to the part similar to the SSTL12 standard because there are no rebiasing resistors on the input signal.

Table 10: Bank 64 Miscellaneous I/O Descriptions

Signal Name	Direction	Pin	I/O	Description
USB_UART_RX	Output	BB20	IO_T3U_N12_64	UART Output to FTDI FT4232 RX input
USB_UART_TX	Input	BF18	IO_L24N_T3U_N11_64	UART Input from FTDI FT4232 TX output
PCIE_PERST_LS	Input	BD21	IO_L23P_T3U_N8_64	Active-Low input from connector to FPGA to detect presence.
QSFP0_MODSELL_LS	Output	BE16	IO_L22N_T3U_N7_DBC_AD0N_64	QSFP, Active-Low, Module Select

Table 10: Bank 64 Miscellaneous I/O Descriptions (cont'd)

Signal Name	Direction	Pin	I/O	Description
QSFP0_RESETL_LS	Output	BE17	IO_L22P_T3U_N6_DBC_AD0P_64	QSFP, Active-Low, Module Reset
QSFP0_MODPRSL_LS	Input	BE20	IO_L21N_T3L_N5_AD8N_64	QSFP, Active-Low, Module Present
QSFP0_INTL_LS	Input	BE21	IO_L21P_T3L_N4_AD8P_64	QSFP, Active-Low, Module Interrupt Output to FPGA
QSFP0_LPMODE_LS	Output	BD18	IO_L20N_T3L_N3_AD1N_64	QSFP, Active-Low, Low Power Mode Enable. Must be High for normal operation.
I2C_MAIN_RESET_B_LS	Input	BF19	IO_L19N_T3L_N1_DBC_AD9N_64	Reset from Satellite controller (SC) to reset I2C MUX
I2C_FPGA_SCL_LS	Input	BF20	IO_L19P_T3L_N0_DBC_AD9P_64	Slave I2C data connection from satellite controller to FPGA.
I2C_FPGA_SDA_LS	Bidirectional	BF17	IO_T2U_N12_64	Slave I2C clock connection from satellite controller to FPGA.
FPGA_TXD_MSP	Output	BB19	IO_L18N_T2U_N11_AD2N_64	Satellite controller CMS UART receive (115200, no parity, 8 bits, 1 stop bit).
FPGA_RXD_MSP	Input	BA19	IO_L18P_T2U_N10_AD2P_64	Satellite controller CMS UART transmit (115200, no parity, 8 bits, 1 stop bit).
STATUS_LED0_FPGA	Input	BC21	IO_L17N_T2U_N9_AD10N_64	LED, Active High, Front Panel orange LED. Can also be driven by SC!
STATUS_LED1_FPGA	Input	BB21	IO_L17P_T2U_N8_AD10P_64	LED, Active High, Front Panel yellow LED. Can also be driven by SC!
STATUS_LED2_FPGA	Input	BA20	IO_L16N_T2U_N7_QBC_AD3N_64	LED, Active High, Front Panel green LED. Can also be driven by SC!
QSFP1_MODSELL_LS	Output	AY20	IO_L16P_T2U_N6_QBC_AD3P_64	QSFP, Active-Low, Module Select.
QSFP1_RESETL_LS	Output	BC18	IO_L15N_T2L_N5_AD11N_64	QSFP, Active-Low, Module Reset.
QSFP1_MODPRSL_LS	Input	BC19	IO_L15P_T2L_N4_AD11P_64	QSFP, Active-Low, Module Present.
QSFP1_INTL_LS	Input	AV21	IO_L14N_T2L_N3_GC_64	QSFP, Active-Low, Module Interrupt Output to FPGA.
QSFP1_LPMODE_LS	Output	AV22	IO_L14P_T2L_N2_GC_64	QSFP, Active-Low, Low Power Mode Enable. Must be High for normal operation.
USER_SI570_CLOCK_N	Input	AV19	IO_L12N_T1U_N11_GC_64	LVDS Reference Clock Input, Default is 156.25 MHz. Negative Signal.
USER_SI570_CLOCK_P	Input	AU19	IO_L12P_T1U_N10_GC_64	LVDS Reference Clock Input, Default is 156.25 MHz. Positive Signal.
SYCLK1_300_N	Input	AW19	IO_L11N_T1U_N9_GC_64	LVDS Reference Clock Input, 300 MHz. Negative Signal.
SYCLK1_300_P	Input	AW20	IO_L11P_T1U_N8_GC_64	LVDS Reference Clock Input, 300 MHz. Positive Signal.
QSFP0_FS0	Output	AT20	IO_L10P_T1U_N6_QBC_AD4P_64	SI5335A Frequency Select Signal. FS[1:0] = 01 -> CLK1A/1B: 156.25 MHz 1.8V LVDS FS[1:0] = 1X -> CLK1A/1B: 161.1328125 MHz 1.8V LVDS

Table 10: Bank 64 Miscellaneous I/O Descriptions (cont'd)

Signal Name	Direction	Pin	I/O	Description
QSFP0_FS1	Output	AU22	IO_L9N_T1L_N5_AD12N_64	SI5335A Frequency Select Signal. FS[1:0] = 01 -> CLK1A/1B: 156.25 MHz 1.8V LVDS FS[1:0] = 1X -> CLK1A/1B: 161.1328125 MHz 1.8V LVDS
QSFP0_REFCLK_RESET	Output	AT22	IO_L9P_T1L_N4_AD12P_64	SI5335A Active-Low, Reset Input to Clock Synthesizer.
QSFP1_REFCLK_RESET	Output	AR21	IO_L8N_T1L_N3_AD5N_64	SI5335A Active-Low, Reset Input to Clock Synthesizer.
QSFP1_FS0	Output	AR22	IO_L8P_T1L_N2_AD5P_64	SI5335A Frequency Select Signal. FS[1:0] = 01 -> CLK1A/1B: 156.25 MHz 1.8V LVDS FS[1:0] = 1X -> CLK1A/1B: 161.1328125 MHz 1.8V LVDS.
QSFP1_FS1	Output	AU20	IO_L7N_T1L_N1_QBC_AD13N_64	SI5335A Frequency Select Signal. FS[1:0] = 01 -> CLK1A/1B: 156.25 MHz 1.8V LVDS FS[1:0] = 1X -> CLK1A/1B: 161.1328125 MHz 1.8V LVDS.
DDR4_RESET_GATE	Output	AU21	IO_L7P_T1L_N0_QBC_AD13P_64	DDR4_RESET_GATE active-High output from FPGA to hold all external DDR4 interfaces in self-refresh. This output disconnects the memory interface reset and holds it in active and pulls the clock enable signal on the memory interfaces.

Known Issues and Limitations

Refer to [Answer Record 71752](#) for known issues with Alveo data center accelerator cards. For card installation support, debugging information, and known issues, see the *Alveo Card Debug Guide* ([XD027](#)). For Technical Support, open a [Support Service Request](#).

Additional Resources and Legal Notices

Finding Additional Documentation

Documentation Portal

The AMD Adaptive Computing Documentation Portal is an online tool that provides robust search and navigation for documentation using your web browser. To access the Documentation Portal, go to <https://docs.xilinx.com>.

Documentation Navigator

Documentation Navigator (DocNav) is an installed tool that provides access to AMD Adaptive Computing documents, videos, and support resources, which you can filter and search to find information. To open DocNav:

- From the AMD Vivado™ IDE, select **Help → Documentation and Tutorials**.
- On Windows, click the **Start** button and select **Xilinx Design Tools → DocNav**.
- At the Linux command prompt, enter `docnav`.

Design Hubs

AMD Design Hubs provide links to documentation organized by design tasks and other topics, which you can use to learn key concepts and address frequently asked questions. To access the Design Hubs:

- In DocNav, click the **Design Hubs View** tab.
- Go to the [Design Hubs](#) webpage.

Note: For more information on DocNav, see the [Documentation Navigator](#) webpage.

Support Resources

For support resources such as Answers, Documentation, Downloads, and Forums, see [Support](#).

References

These documents provide supplemental material useful with this guide:

Product Websites

The most up-to-date information related to the AMD Alveo™ U200/U250 card and documentation is available on the following websites:

1. [Alveo U200 Data Center Accelerator Card](#)
2. [Alveo U250 Data Center Accelerator Card](#)
3. Alveo Data Center Accelerator Card - Known Issues and General Information ([AR 71752](#))

Supplemental Documents

The following AMD document provide supplemental material useful with this guide.

- *Alveo U200 and U250 Data Center Accelerator Cards Data Sheet* ([DS962](#))
- *Vitis Unified Software Platform Documentation* ([UG1416](#))
- *UltraFast Design Methodology Guide for FPGAs and SoCs* ([UG949](#))
- *Vivado Design Suite User Guide: Designing IP Subsystems using IP Integrator* ([UG994](#))
- *Getting Started with Alveo Data Center Accelerator Cards* ([UG1301](#))
- *Alveo Data Center Accelerator Card Platforms User Guide* ([UG1120](#))
- *UltraScale Architecture Configuration User Guide* ([UG570](#))
- *Vivado Design Suite User Guide: Programming and Debugging* ([UG908](#))
- *Card Management Solution Subsystem Product Guide* ([PG348](#))
- *Alveo Card Debug Guide* ([XD027](#))

Additional Links

The following links provide supplemental material useful with this guide.

- Xilinx, Inc: <https://www.xilinx.com>
- [Answer Record 71757](#)

- [Answer Record 71752](#)
- Micron Technology: <http://www.micron.com>
(MTA18ASF2G72PZ-2G3B1IG, MT25QU01GBB8E12-OSIT)
- *Si5394 Data Sheet*: <https://www.silabs.com/documents/public/data-sheets/si5395-94-92-a-datasheet.pdf>
- Future Technology Devices International, Ltd.: <http://www.ftdichip.com>
(FT4232HQ)
- SFP-DD module: [SFP-DD Specification](#)

Revision History

The following table shows the revision history for this document.

Section	Revision Summary
06/13/2023 Version 1.1	
MCS File Generation and Alveo Card Programming	• Added step to address configuration of the memory device. • Corrected write_cfgmem Parameter Settings size.
05/05/2023 Version 1.0	
Initial release.	N/A

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